Miles Thompson

AI / Data Science / Developer

Wellington, New Zealand • +64 21-539-843 • utunga@gmail.com • goodbright.nz

AI Engineer, programmer, and sometimes tech lead. I have pretrained and deployed foundation models from scratch. Recent work on runtime agentic AI governance. Two startups I co-founded were acquired. I've presented at NeurIPS[1], ICML[2] and AAAI[3]

EXPERIENCE HIGHLIGHTS

CHURPRIMO 2026 –

Contract CTO / Agentic Engineering

- Built cutting-edge runtime governance for agentic AI governance. Developed business model.
- Presented our research in the Multi Agent Systems track at AAAI Spring Symposium Series, San Francisco[3]

Te Hiku Media, New Zealand 2020 – 2026

Machine Learning Engineer / AI Lead Dev

- Pretrained foundation language models, achieving best in class performance for our chosen languages. As technical lead helped build the team, installed custom GPU hardware, built ai-ops for training and dev-ops to deploy our 100% custom AI models at cloud scale using PyTorch, Python, FastAPI, Kubernetes
- The contributions of our small research lab were recognised when *our founder was awarded the* ***Time100 most influential people in AI list****, in 2024*

Chatterize / Speakia, New Zealand / China 2019 – 2021

Contract CTO

- Brainstormed strategy and pitching, chose the AI platform, hired development team and led early development and deployment of our AI based language learning app
- After achieving hockey stick shaped user growth in the complex Chinese mobile app market, and a rebrand as "Speakia", we were ***acquired by EduMaxi, May 2022***

CreditSights, New York, USA 2005 – 2019

Chief Technology Officer (CTO)

- As first CTO, then senior technical architect, helped build the early risk tools, product stack and financial modeling capability of this well-known New York financial research company
- My contributions are reflected in strong growth during our startup phase, while I was CTO, as well as solid revenue growth to $50m+ during my time there overall. CreditSights was ***acquired by Fitch Ratings in 2020***

EDUCATION

BSc Hons I, Maths, University of Otago, New Zealand

First class honours — Mathematics (aka "Summa cum laude").

BA, Sociology, University of Canterbury, New Zealand

Bachelor of Arts — Sociology

SKILLS (HIGHLIGHTS)

Agentic AI: OpenClaw, Hermes, SwarmMap, Model Introspection, Custom Agentic Harness
Cloud / AI / Dev Ops: AWS, Azure, OpenStack, Kubernetes, Terraform, Azure, FastAPI, Bash, Linux

Data science / Machine Learning: Pretraining Transformer and Conformer models, Python, PyTorch, Tensorflow, NeMo, Pandas, Numpy, GANs, Bayesian modeling, Financial Risk Models
Web development: ReactJS, NextJS, VueJS, Angular, Firebase, Wordpress, Apache, Network Ops,
Product development: UX, Design, Figma, Lean methodology, Customer discovery, Startup founder
Business development: Customer discovery, startup pitches, detailed business plans, fund raising
Preferred programming languages: Python, C#, Infrastructure as code (Terraform/YAML etc)

EXPERIENCE (HIGHLIGHTS)

'Connie' - Speech recognition and synthesis for Māori and NZ English 2021 - 2024
Machine learning engineer, AI lead dev

- Led development of a bilingual, AI language model, from scratch *without using any dubiously obtained data.*
- Managed, architected and built cloud based kubernetes cluster to deploy highly scalable AI API at https://papareo.io
- Achieved *higher accuracy than any other vendor* for our target languages, including in the NZ English subset.
- I presented our research at *ICML 2023* [2]*: The Creation of a Bilingual Māori-New Zealand English ASR*
- Our benchmarks and process were published at *NeurIPS 2024* [4]*: Fit for our purpose, not yours: Benchmark for a low-resource, Indigenous language*

'Rongo' - On device pronunciation feedback 2021
Machine learning engineer, individual contributor

- Conceived and coded an on-device "pronunciation feedback" model using internal values of neural model
- Presented research as a poster at *NeurIPS 2020* [1]*: Scoring pronunciation accuracy via close introspection of a speech recognition recurrent neural network*
- Achieved first in education and fifth overall in the NZ IPhone App market for our 2021 app https://rongo.app

Resilience App: Bayesian simulation for communications in a disaster 2019 - 2021
Mathematical modeler, project founder, open source contributor

- Authored a white paper and code simulating the impact of prioritized messaging in a disaster situation
- Identified ways that resilient communications can transform aid and finance to the developing world and reliable communication during disaster or crisis situations
- Based on this work, co-founded and helped organise hundreds of volunteers to deliver food via an app in LA during the early months of the COVID pandemic, resulting in the 'resilience app' open source project
- Thousands of boxes of donated food were delivered and hundreds of volunteers contributed to the app

Reserve Bank of New Zealand: Empowering financial modeling 2019
Contract programmer

- Optimized performance of mission critical Reserve Bank database and ETL process for acquiring, aggregating, filtering and analyzing data from reporting financial institutions
- In a short contract succeeded in delivering a five fold performance improvement and other user experience improvements over an already heavily optimized system. Garnered a lot of appreciation from banking analysts for making their jobs easier and more productive

RealityVirtual: Virtual reality assets from photographs 2019
Machine learning engineer

- Trained a generative adversarial network (GAN) that generates the digital assets necessary for high fidelity virtual reality from simple photographs (texture maps, bump maps)
- Deployed a working demo of our model as an interactive web tool. Received positive investor interest

CreditSights 2005 - 2019

Technical architect, 'all of market' daily bond pricing system

- Re-architected and implemented an ETL process that ingests all of market daily bond pricing. Optimized a process that used to run once a week so that it could run in a few hours while simultaneously building a faster more accessible system, empowering analysts to make easier queries and calculations.

Technical architect, across multiple projects

- Technical architecture, team lead and hands on implementation for our flagship research products, delivering quantitative and qualitative content in an integrated view across multiple projects. Development of systems to capture and clean data from 10K, 10Qs and other data sources into an integrated view. Success is reflected in steady revenue growth and multiple awards for the quality of our research.

The Great Outdoors: Algorithmic theater before generative AI was a thing 2016 - 2017

Machine learning engineer, member of troupe

- Created a machine learning model to realise the theatrical vision of renowned director Annie Dorsen and her troupe, creating an ever changing theater piece incorporating the current zeitgeist based on 'found text'
- Built a custom data pipeline that sifts, classifies and selects text from the internet based on measures of entropy and other semantic factors and deployed it into a form factor that could travel with the troupe
- *The Great Outdoors opened to good reviews in The Netherlands, Paris, Berlin, New York, and Boston (review)*

Hashmapd: Applying 'deep learning' to social media, in the early days of deep learning 2010 - 2011

Machine learning engineer, startup founder

- Along with my co-founders Dr Abraham and Dr Frean, used early deep learning models to process social media text using 'semantic hashing'
- Built a product demo, explored potential investment. Made all our code open source in 2014

Medigenesis: medical collaborative filtering and cluster analysis system 2000 - 2001, 2008

Technical architect, technology lead, co-inventor

- Working with digital medicine pioneer Dr Sidney Baker (MD) led a team of eight developers to realise his vision for applying statistical techniques to medical data. The system recommends treatments that work for 'others like you' based on medical profile, especially effective for complex and chronic diseases
- Along with Dr Baker, was awarded a patent for work completed in 2001. Patent granted 2009
- Later, in 2008 helped secure funding to pivot to a non-profit structure and relaunched the project as 'Autism 360'

REFEREES

Contact details available on request

Keoni Mahelona: Chief Technology Officer, Te Hiku Media
Keoni was my direct manager at Te Hiku Media, 2021- 2026

Greg Perrott: Principal Solutions Architect at Reserve Bank of New Zealand
Greg was my direct manager at RBNZ in 2019

Shaun McCarthy: Research Engineering Leader, Object Indexing @ Reality Labs (Meta)
Shaun was my direct manager at CreditSights 2008 - 2013

Judith Chinitz: MS, MS, CNC Co-Founder Biome Restoration Ltd
I worked for Judy as a software vendor on Medigenesis (2000 - 2001) and later on Autism360.org (2008)

REFERENCES

[1] *Moses, C. & Thompson, M. Poster, NeurIPS 2020*

[2] *Stevens, L. & Thompson, M. The Creation of a Bilingual Māori–NZ English ASR. Indigenous in AI Workshop, ICML 2023, Honolulu. icml.cc/virtual/2023/29889*

[3] *Sughero, K. L. & Thompson, M. Presentation and live demo, Multi-Agent System Safety and Teamwork (MASST) Symposium, AAAI Spring Symposium Series 2026. (programme)*

[4] *Duncan, S., Leoni, G., Steven, L., Mahelona, K. & Jones, P.-L. Fit for our purpose, not yours. NeurIPS 2024, Datasets & Benchmarks Track. Te Hiku Media. proceedings.neurips.cc/paper*